

# Approach to Patients and Families with Genetic Disorders

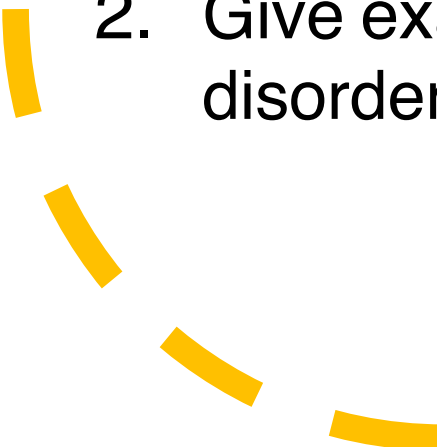
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2026





# Objectives:

At the end of the lecture, you should be able to:

1. Discuss the clinical genetic approach to patients and families with genetic disorders.
  2. Give examples of approaches to patients and families with genetic disorders.
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# Outline



Multifactorial disorders



Single gene disorders



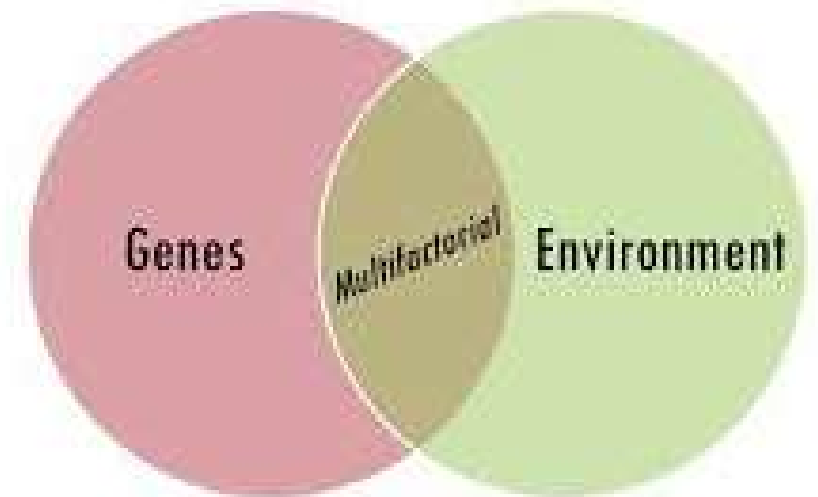
Chromosomal abnormalities

# Multifactorial Disorders

? Multifactorial inheritance describes the action of multiple **genes** that interact with one another and with **environmental** factors to lead to a phenotype

→ interaction of genetic and environmental factors

? They do not follow a simple Mendelian transmission pattern (non-Mendelian familial aggregation)



# Multifactorial Causation

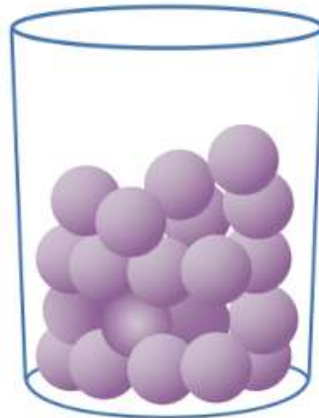


Genetic Factors



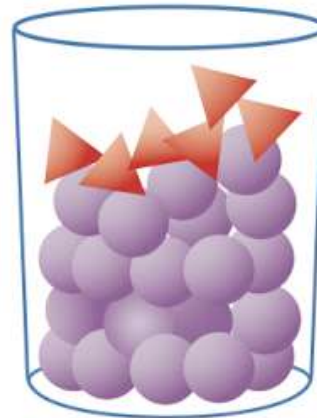
Environmental Factors

Genetic Background



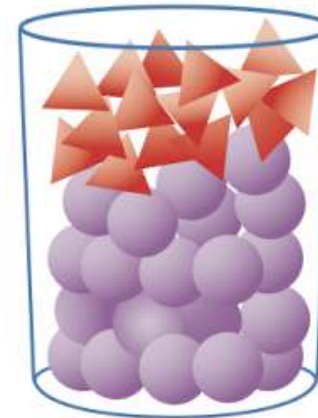
Not enough factors to cause the condition

Genetic Background +  
Some Environmental Factors



Higher risk,  
but still not enough factors  
to cause the condition

Genetic Background +  
More Environmental Factors



Genetic and environmental  
factors combine to cause  
the condition



# Multifactorial Disorders

- ❓ Multifactorial disorders are believed to constitute most common traits as well as majority of the isolated "non-syndromic" malformations
- ❓ Recurrence risks to relatives for multifactorial disorders are influenced by:
  - The disease severity
  - The degree of relationship to the index case
  - The number of affected close relatives
  - If there is a higher incidence in one sex
  - The sex of the index case

## Box 10.1 Disorders That Show Multifactorial Inheritance

### Congenital Malformations

Cleft lip/palate  
Congenital dislocation of the hip  
Congenital heart defects  
Neural tube defects  
Pyloric stenosis  
Talipes

### Acquired Diseases of Childhood and Adult Life

Asthma  
Autism  
Diabetes mellitus  
Epilepsy  
Glaucoma  
Hypertension  
Inflammatory bowel disease (Crohn disease and ulcerative colitis)  
Ischemic heart disease  
Ischemic stroke  
Bipolar disorder  
Multiple sclerosis  
Parkinson disease  
Psoriasis  
Rheumatoid arthritis  
Schizophrenia

# Neural Tube Defects (NTD)

- Neural tube closes around 4<sup>th</sup> week of gestation
- NTD result from defective closure of the developing neural tube;
  - A defect at the **upper** end results in **anencephaly** or **encephalocele**
  - A defect at the **lower** end leads to a spinal lesion such as a lumbosacral **meningocele** or **myelomeningocele**
- Can be **isolated**, or due to **chromosomal** abnormalities or part of a **syndrome**
- Most are isolated malformations in otherwise normal infants and show multifactorial inheritance
- Periconceptional folic acid supplementation reduces the risk of recurrence by 70% when a woman has had one affected child
- NTD can be detected prenatally by elevated amniotic alpha fetoprotein (AFP) level and US





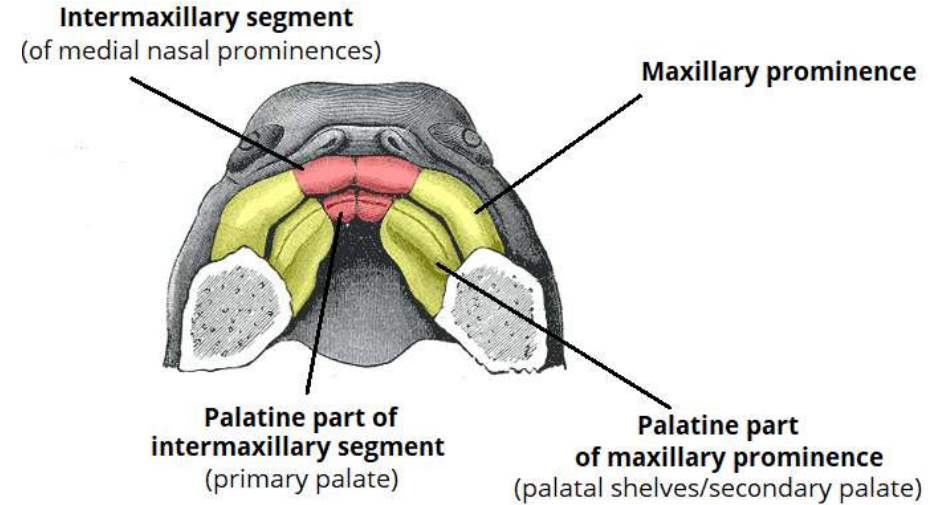
| Relationship of affected individual to 'at risk' pregnancy | Recurrence risk (%) |
|------------------------------------------------------------|---------------------|
| One sibling                                                | 2-5%                |
| Two siblings                                               | 10-12%              |
| One 2 <sup>nd</sup> degree relative                        | 1-2%                |
| One 3 <sup>rd</sup> degree relative                        | 0.5-1%              |
| One parent                                                 | 4%                  |

## Neural Tube Defects (NTD)

Source\* Oxford Desk Reference  
Clinical Genetics by Helen V. Firth  
and Jane A. Hurst



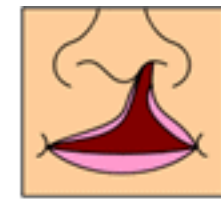
# Cleft Lip (CL) and Cleft Palate (CP)



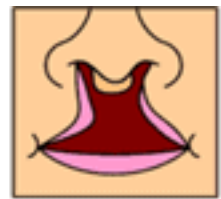
- A cleft refers to a gap/split in the upper lip or palate
- It results from a defect during development of face and palate



Unilateral incomplete



Unilateral complete



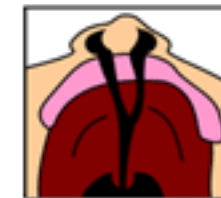
Bilateral complete



Incomplete cleft palate



Unilateral complete lip and palate



Bilateral complete lip and palate

# Cleft Lip (CL) and Cleft Palate (CP)

Source\* Oxford Desk Reference Clinical Genetics by Helen V. Firth and Jane A. Hurst

## Recurrence risk for isolated CL and isolated CL/P

| Relationship to index case | Recurrence risk (%) |
|----------------------------|---------------------|
| Sibling unilateral CL      | 2-3                 |
| Sibling unilateral CL/P    | 4                   |
| Sibling bilateral CL/P     | 5-6                 |
| Two affected siblings      | 10                  |
| Affected parent            | 4                   |

## Recurrence risk for isolated CP

| Relationship to index case | Recurrence risk (%) |
|----------------------------|---------------------|
| Sibling                    | 2-3                 |
| Affected parent            | 4                   |

**Table 8.7** Empiric Recurrence Risks for Common Multifactorial Disorders

| Disorder                          | Incidence (Per 1000) | Sex Ratio (M : F) | Unaffected Parents Having a Second Affected Child (%) | Affected Parents Having an Affected Child (%) |
|-----------------------------------|----------------------|-------------------|-------------------------------------------------------|-----------------------------------------------|
| Cleft lip $\pm$ cleft palate      | 1–2                  | 3 : 2             | 4                                                     | 4                                             |
| Clubfoot (talipes)                | 1–2                  | 2 : 1             | 3                                                     | 3                                             |
| Congenital heart defect           | 8                    | 1 : 1             | 1–4                                                   | 2 (father affected)<br>6 (mother affected)    |
| Congenital dislocation of the hip | 1                    | 1 : 6             | 6                                                     | 12                                            |
| Hypospadias (in males)            | 2                    | —                 | 10                                                    | 10                                            |
| Manic depression                  | 4                    | 2 : 3             | 10–15                                                 | 10–15                                         |
| <b>Neural Tube Defect</b>         |                      |                   |                                                       |                                               |
| Anencephaly                       | 1.5                  | 1 : 2             | 4–5                                                   | —                                             |
| Spina bifida                      | 2.5                  | 2 : 3             | 4–5                                                   | 4                                             |
| <b>Pyloric Stenosis</b>           |                      |                   |                                                       |                                               |
| Male index                        | 2.5                  | —                 | 2                                                     | 4                                             |
| Female index                      | 0.5                  | —                 | 10                                                    | 17                                            |
| Schizophrenia                     | 10                   | 1 : 1             | 10                                                    | 14                                            |

For many of the multifactorial disorders, empirical risks have been derived based on large epidemiological family studies, so that it is usually possible to provide the parents of an affected child with a clear indication of the likelihood that a future child will be similarly affected.

# Single Gene Disorders

- Account for up to 10% of all congenital abnormalities
- Some are isolated, affecting only one organ or system, while others cause multiple congenital abnormality syndromes impacting various, seemingly unrelated organ systems.
- They run in families and can be dominant or recessive, and autosomal or sex-linked
- Pedigree analyses are very useful for determining the inheritance pattern of single-gene disorder

**Table 16.4** Congenital abnormalities that can be caused by single-gene defects

| Inheritance Abnormalities     |       |                                                      |
|-------------------------------|-------|------------------------------------------------------|
| <u>Isolated</u>               |       |                                                      |
| <u>CENTRAL NERVOUS SYSTEM</u> |       |                                                      |
| Hydrocephalus                 | XR    |                                                      |
| Megalencephaly                | AD    |                                                      |
| Microcephaly                  | AD/AR |                                                      |
| <u>OCULAR</u>                 |       |                                                      |
| Aniridia                      | AD    |                                                      |
| Cataracts                     | AD/AR |                                                      |
| Microphthalmia                | AD/AR |                                                      |
| <u>LIMB</u>                   |       |                                                      |
| Brachydactyly                 | AD    |                                                      |
| Ectrodactyly                  | AD/AR |                                                      |
| Polydactyly                   | AD    |                                                      |
| <u>OTHER</u>                  |       |                                                      |
| Infantile polycystic kidneys  | AR    |                                                      |
| <u>Syndromes</u>              |       |                                                      |
| Apert                         | AD    | Craniosynostosis, syndactyly                         |
| EEC                           | AD    | Ectodermal dysplasia, ectrodactyly, cleft lip/palate |
| Meckel                        | AR    | Encephalocele, polydactyly, polycystic kidneys       |
| Roberts                       | AR    | Cleft lip/palate, phocomelia                         |
| Van der Woude                 | AD    | Cleft lip/palate, lip pits                           |



**FIG. 16.10** Appearance of the feet in a child with ectrodactyly.



Essential  
Information  
on Single-  
Gene  
Disorders

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Responsible gene

---

Cytogenetic locus

---

Inheritance

---

Disease mechanism

---

Clinical features

---

Diagnosis (clinical and molecular tests)

---

Management/treatment

# Autosomal Dominant Disorders

- Neurofibromatosis type 1
- Marfan syndrome
- Achondroplasia
- Noonan syndrome





# Marfan Syndrome

## Responsible gene:

*FBN1* (encodes for fibrillin-1 protein, extracellular matrix protein that contributes to microfibrils that are found in both elastic and non-elastic tissues)

## Cytogenetic locus:

15q21

## Inheritance:

AD; 25% de novo

## Disease mechanism:

Dominant negative pathogenic variants, resulting in < 35% of the expected amount of fibrillin-1 in the extracellular matrix.



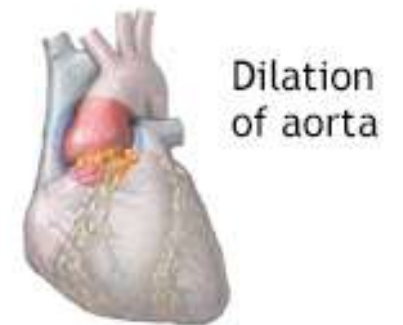


# Marfan Syndrome

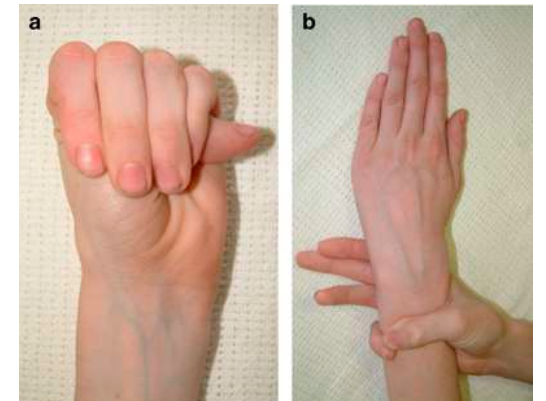
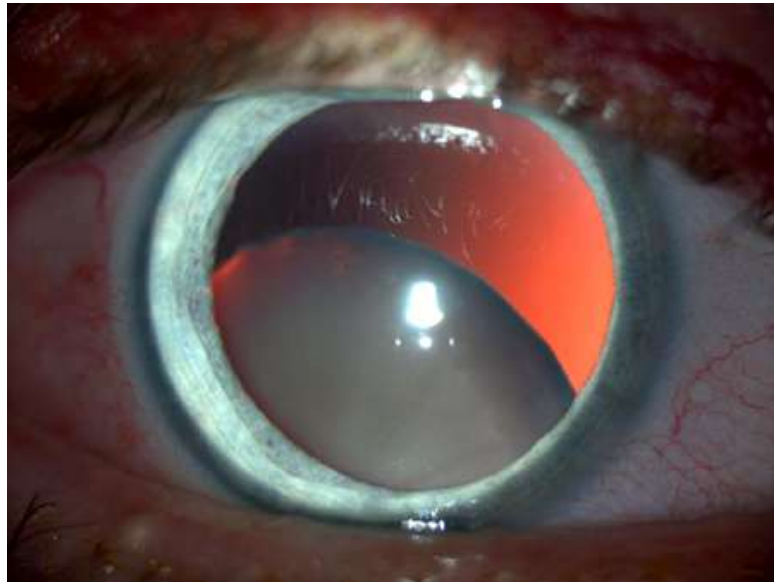
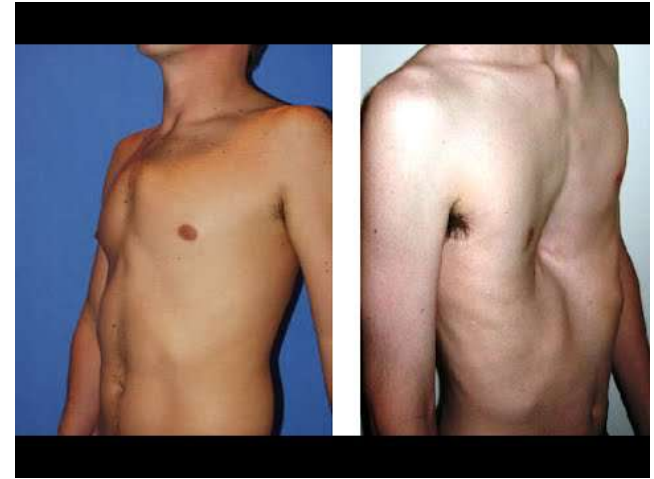
## Clinical Features:

Systemic disorder of connective tissue that involves ocular, skeletal and cardiovascular systems (pleiotropy)

- **CV:** aortic root dilation, aortic aneurysm/dilation of the ascending aorta, mitral valve prolapse
- **Skeletal:** pectus carinatum or excavatum, reduced upper/lower segment, arm span/height > 1.05, scoliosis, pes planus (flat feet), joint laxity, and arachnodactyly
- **Eye:** ectopia lentis (anterior displacement of the lens)
- **Skin:** skin striae (stretch marks) and hernia

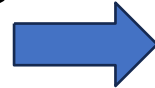


ADAM.



# Marfan Syndrome

**Diagnosis:** Based on the revised Ghent criteria



**Table 19.5** Revised Ghent criteria for making a diagnosis of Marfan syndrome

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <u>For a Diagnosis of Marfan Syndrome (MFS) (With No Family History):</u>                                    |
| (1) Aortic root dilatation (Z score $\geq 2$ ) plus ectopia lentis                                           |
| (2) Aortic root dilatation (Z score $\geq 2$ ) plus pathogenic <i>FBN1</i> mutation                          |
| (3) Aortic root dilatation (Z score $\geq 2$ ) plus Systemic Score $\geq 7$ points (below)                   |
| (4) Ectopia lentis plus pathogenic <i>FBN1</i> mutation with known aortic root diameter                      |
| <u>If Family History (FH) Present:</u>                                                                       |
| (5) Ectopia lentis plus FH of MFS, as defined above                                                          |
| (6) Systemic Score ( $\geq 7$ points) plus FH of MFS                                                         |
| (7) Aortic root dilatation (Z score $\geq 2$ above 20 years old; $\geq 3$ below 20 years old) plus FH of MFS |

## Management:

- Symptom management by a multidisciplinary team.
- Aortic complication prevention with  $\beta$ -adrenergic blocker and angiotensin-II receptor blocker (ARBs)
- Regular surveillance (eye, spine, cardiac) and avoidance of risky activities/meds.

# Marfan Syndrome Systemic Scoring

- Features other than aortic root enlargement or ectopia lentis are grouped to derive a “systemic score” that can contribute to diagnosis
- A score of 7 and above considered positive

<https://www.marfan.org/dx/score>








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## CALCULATION OF SYSTEMIC SCORE

Clinical manifestations of MFS in other organ systems were critically evaluated for their specificity and diagnostic utility based on expert opinion and the available literature. Several of the “minor” criteria from the old Ghent nosology were eliminated, but the most selective systemic features were included in the “systemic score”.

| Feature                                                             | Value | Click to include         |
|---------------------------------------------------------------------|-------|--------------------------|
| Wrist AND thumb sign                                                | + 3   | <input type="checkbox"/> |
| Wrist OR thumb sign                                                 | + 1   | <input type="checkbox"/> |
| Pectus Carinatum Deformity                                          | + 2   | <input type="checkbox"/> |
| Pectus Excavatum or Chest Asymmetry                                 | + 1   | <input type="checkbox"/> |
| Hindfoot Deformity                                                  | + 2   | <input type="checkbox"/> |
| Plain Flat Foot                                                     | + 1   | <input type="checkbox"/> |
| Spontaneous Pneumothorax                                            | + 2   | <input type="checkbox"/> |
| Dural Ectasia                                                       | + 2   | <input type="checkbox"/> |
| Protucio Acetabulae                                                 | + 2   | <input type="checkbox"/> |
| Scoliosis or Thoracolumbar Kyphosis                                 | + 1   | <input type="checkbox"/> |
| Reduced Elbow Extension                                             | + 1   | <input type="checkbox"/> |
| 3 of 5 Facial Features                                              | + 1   | <input type="checkbox"/> |
| Skin Striae                                                         | + 1   | <input type="checkbox"/> |
| Severe Myopia                                                       | + 1   | <input type="checkbox"/> |
| Mitral Valve Prolapse                                               | + 1   | <input type="checkbox"/> |
| Reduced Upper Segment / Lower Segment & Increased Arm span / Height | + 0   | Open to calculate        |

\*A score of  $\geq 7$  is considered a positive systemic score.



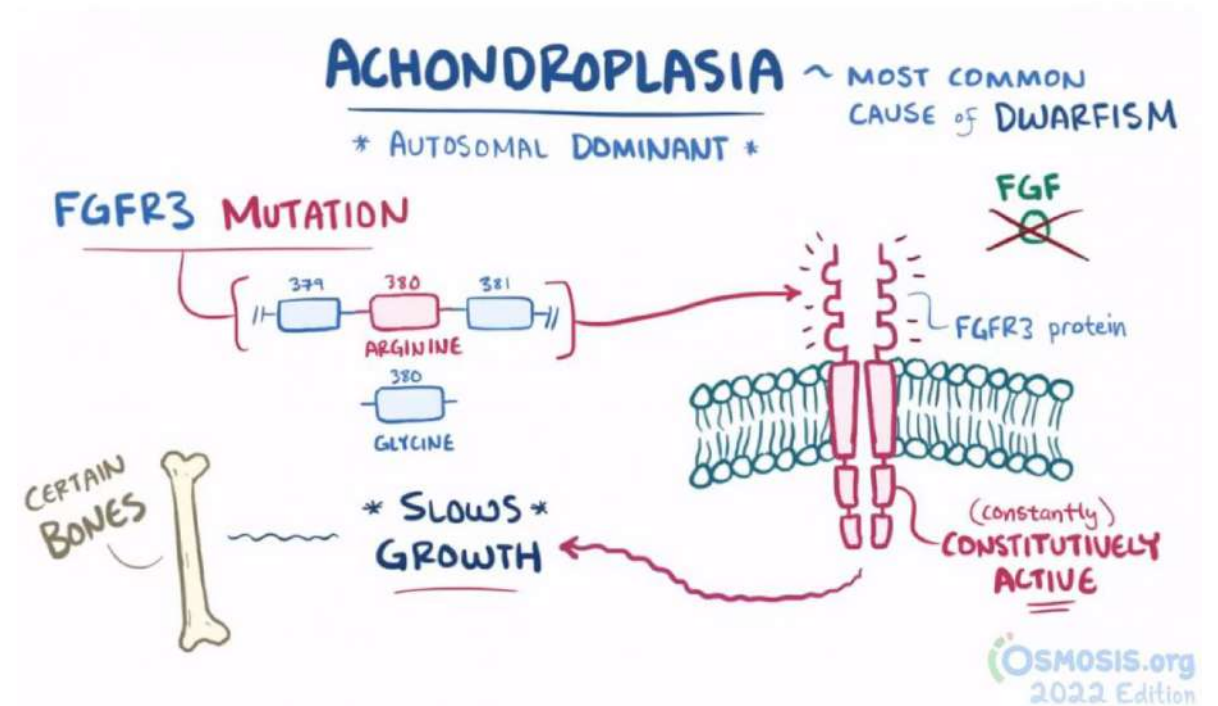
# Achondroplasia

**Responsible gene:** *FGFR3* (encodes fibroblast growth factor receptor 3 protein, a negative regulator of bone growth, which plays a role in slowing of proliferation and differentiation of chondrocytes)

**Cytogenetic locus:** 4p16.3

**Inheritance:** AD; 80% de novo

**Disease mechanism:** Gain of function pathogenic variant (mostly p.Gly380Arg) causes activation of *FGFR3* gene → inhibition of chondrocyte proliferation and differentiation



# Achondroplasia

## Clinical features:

- Most common cause of disproportionate short stature
  - Short stature, rhizomelic (proximal) shortening of limbs, trident hand
  - Frontal bossing, midface hypoplasia, macrocephaly
  - OSA, spinal cord compression
  - Intelligence and life expectancy are normal
- **Management:**
  - Symptomatic
  - Vosoritide (CNP analog, inhibit FGFR3 signaling) can enhance height in children until growth plates close



**FIG. 9.21** Two young patients with achondroplasia. (A) An infant showing typical frontal bossing and excess skin folds as a result of short long bones. (B) An older child who is part of a family with achondroplasia.

# Noonan Syndrome

## Responsible genes:

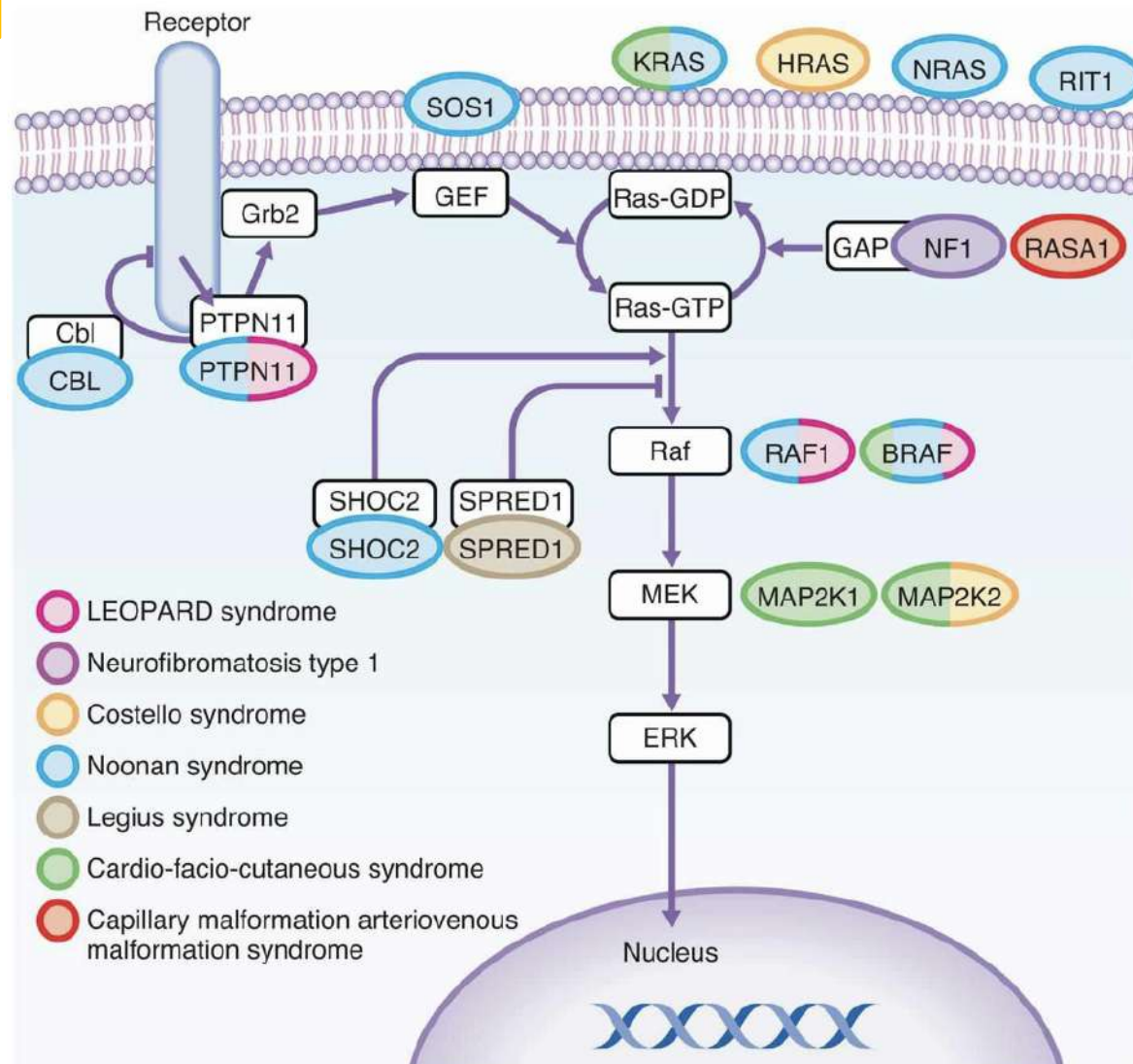
Genes belong to the RAS- MAPK pathway; mainly *PTPN11* (50%), *SOS1* (10%), and others

## Inheritance:

AD, mostly *de novo*

## Disease Mechanism:

Gain of function variant that leads to activation of the Ras-MAPK pathway



# Noonan Syndrome

## Clinical Features:

- Characteristic facial features (hypertelorism, down-slanting palpebral fissures, and low-set/posteriorly rotated ears)
- Short stature
- Congenital heart disease (pulmonary valve stenosis and hypertrophic cardiomyopathy)
- DD/ID/learning difficulties
- Other: neck webbing, pectus deformity, genitourinary abnormalities, lymphedema, bleeding disorders

## Clinical Tests:

Echocardiogram, renal ultrasound, bleeding studies

## Molecular Tests:

*PTPN11* sequencing (50%), *SOS1* sequencing (10%)

## Treatment:

Symptomatic, early intervention, growth hormone replacement





# Neurofibromatosis Type 1

**Responsible gene:** *NF1* (encodes for neurofibromin protein, which controls cellular proliferation and acting as a tumor suppressor)

**Cytogenetic locus:** 17q11

**Inheritance:** AD; 50% de novo

## Clinical Features:

- Multiple café au lait spots (pigmented skin lesions)
- Multiple neurofibromas (benign tumors of the skin)
- Axillary or inguinal freckling
- Optic nerve glioma
- Lisch nodules (pigmented hamartoma of the iris)
- Mild developmental delay and learning difficulties
- Variable expressivity between family members

**Disease Mechanism:** Loss of function pathogenic variant leads to activation of the RAS- MAPK pathway



# Autosomal Recessive Disorders

- Spinal muscular atrophy
- Cystic fibrosis



# SPINAL MUSCULAR ATROPHY

**Responsible genes:** *SMN1*, *SMN2* (encode for survival motor neuron protein 1&2)

**Cytogenetic loci:** 5q13

**Clinical Features:** Characterized by muscle weakness (symmetric, proximal > distal, and progressive) and atrophy resulting from progressive degeneration and loss of the anterior horn cells in the spinal cord (i.e., lower motor neurons)

- SMA I onset < 6mo, muscle weakness, tongue fasciculations, absent deep tendon reflexes (DTRs)
- SMA II muscle weakness onset > 6 months, finger trembling, low tone, absent DTRs
- SMA III weakness leads to frequent falls or trouble with stairs, onset >12-18 months, proximal weakness (legs>arms)
- SMA IV adult onset

**Clinical Tests:** Motor unit disease on electromyogram

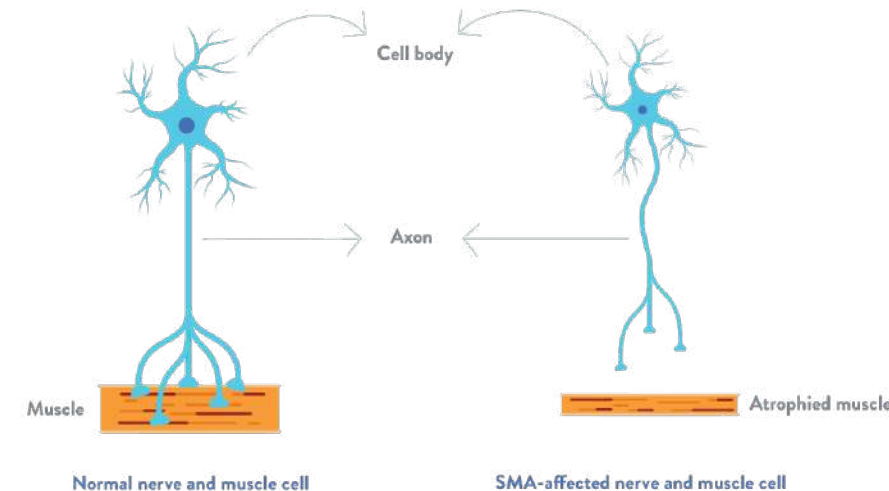
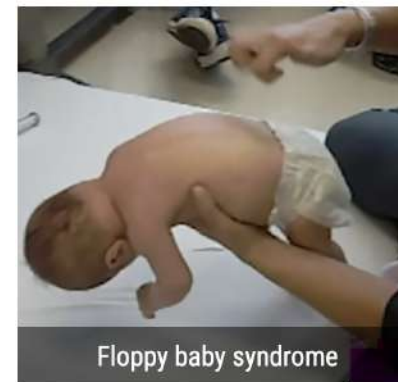


Image retrieved from:  
[https://www.togetherinsma.com/en\\_us/home/introduction-to-sma/smn1-gene.html](https://www.togetherinsma.com/en_us/home/introduction-to-sma/smn1-gene.html)



Floppy baby syndrome

Image from Videos on File. AveXis, Inc., 2018.<sup>7</sup>



Frog-leg position

Image used with permission from CureSMA.<sup>8</sup>



Impaired head control

Image reprinted by permission from Elsevier. From: Sumner CJ, Pauskin S, Ko CP, eds. *Spinal muscular atrophy*. Amsterdam, the Netherlands: Elsevier; 2017.<sup>9</sup>

Images retrieved from: <https://www.zolgensma-hcp.com/understanding-sma/what-is-sma/>

# SPINAL MUSCULAR ATROPHY

## Molecular Tests:

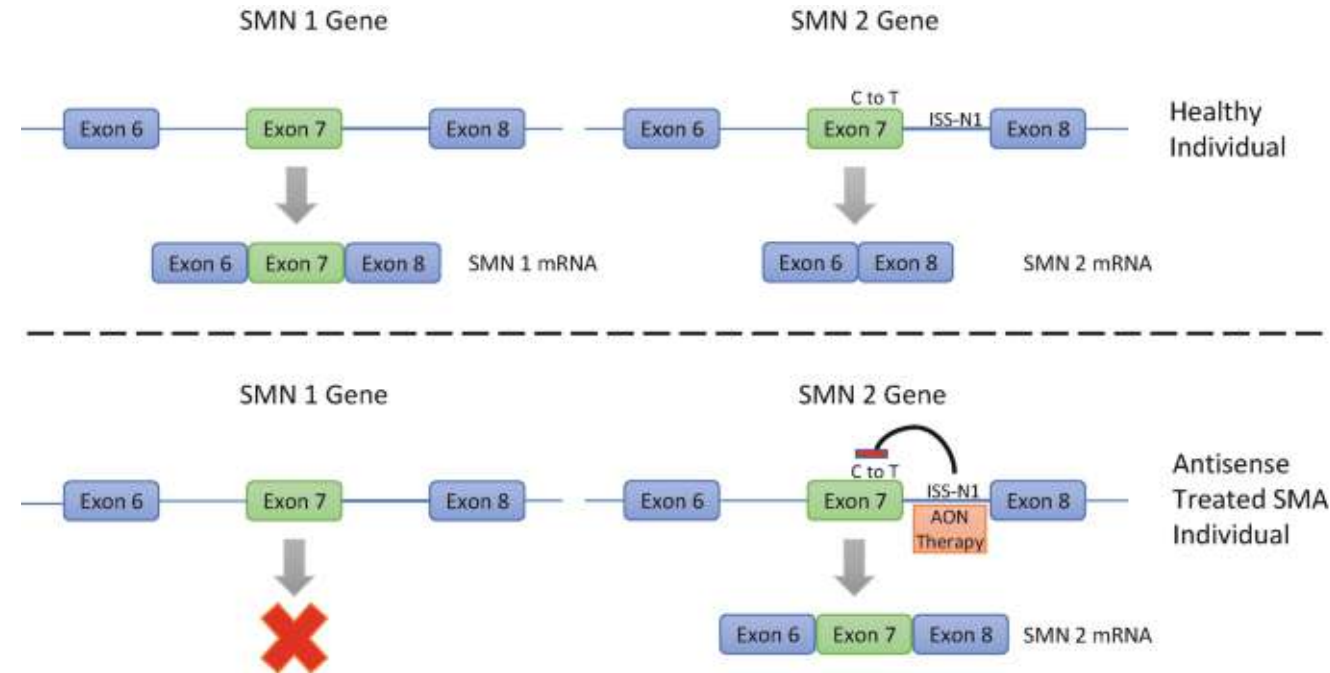
- Biallelic pathogenic variant in *SMN1* (deletion of *SMN1* exon 7 deletion (95-98%), point mutations (2-5%))

## Disease Mechanism:

- Caused by loss of *SMN1* because *SMN2* cannot fully compensate for loss of *SMN1*-produced protein

## Treatment:

- Supportive care
- Spinraza® (antisense oligonucleotide for all types of SMA)
- Zolgensma® (SMN1 gene replacement therapy for SMA I)



Goodkey, K., Aslesh, T., Maruyama, R., & Yokota, T. (2018). Nusinersen in the treatment of spinal muscular atrophy. *Exon skipping and inclusion therapies: Methods and protocols*, 69-76.

# Cystic Fibrosis

## Responsible gene:

*CFTR* (encodes a cystic fibrosis transmembrane conductance regulator protein, which maintains the balance of salt and water)

**Cytogenetic locus:** 7q31

## Clinical Features:

- Chronic pulmonary disease due to chronic airway infection and chronic sinusitis
- Failure to thrive and malnutrition as result of chronic diarrhea and malabsorption due to pancreatic insufficiency, meconium ileus
- Male infertility due to azoospermia and congenital bilateral absence of the vas deferens (CBAVD)

## Clinical Tests:

- Sweat chloride test (elevated sweat chloride levels  $> 160$  mEq/L)

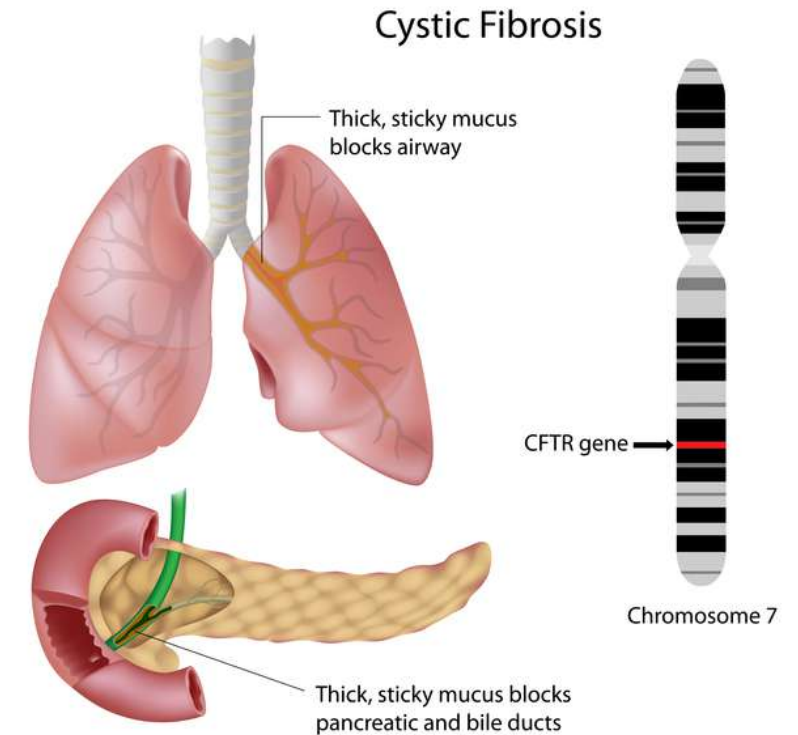


Image retrieved from: <https://medlineplus.gov/genetics/condition/cystic-fibrosis/>



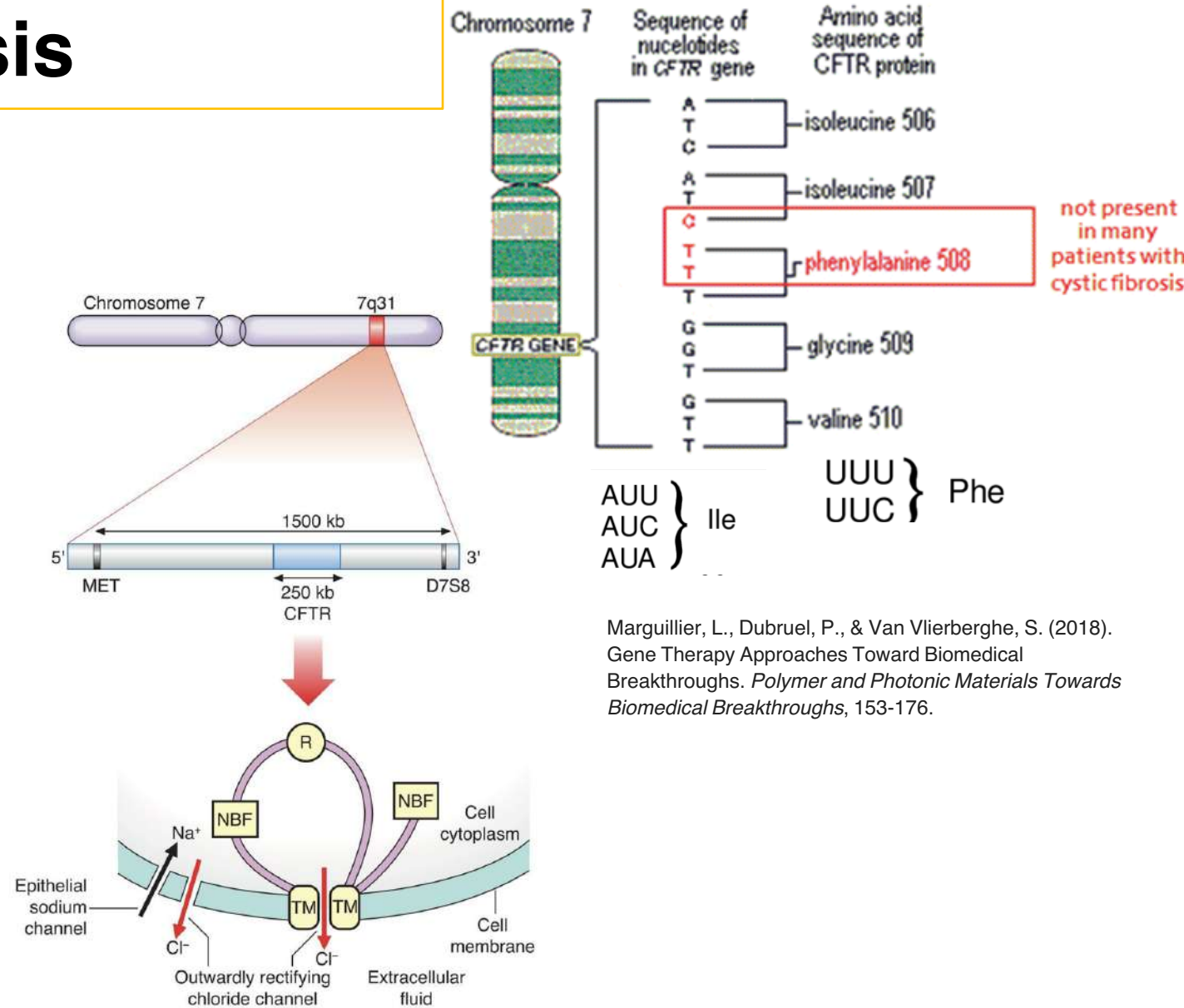
# Cystic Fibrosis

## Molecular Tests:

- Common mutation deltaF508 (a deletion of three base pairs at the 508th codon which results in the loss of a phenylalanine residue) accounts for approximately 70% of all mutations in *CFTR* depending upon ethnic group.

## Disease Mechanism:

- LOF pathogenic variant in the *CFTR* gene disrupt the function of the **chloride channels** preventing them from regulating the flow of chloride ions and water across cell membranes.
- Without Cl, Na will not form salt, and water will not be attracted.
- This leads to excessively thick secretions and tissue damage.



Marguillier, L., Dubruel, P., & Van Vlierberghe, S. (2018). Gene Therapy Approaches Toward Biomedical Breakthroughs. *Polymer and Photonic Materials Towards Biomedical Breakthroughs*, 153-176.



# X-Linked Disorders

- Duchenne muscular dystrophy
- Fragile X syndrome (*covered in Non-Mendelian inheritance lecture*)



# Duchenne Muscular Dystrophy

**Responsible gene:** *DMD* (Dystrophin)

**Cytogenetic locus:** Xp21

**Inheritance:** X-linked recessive

**Clinical features:**

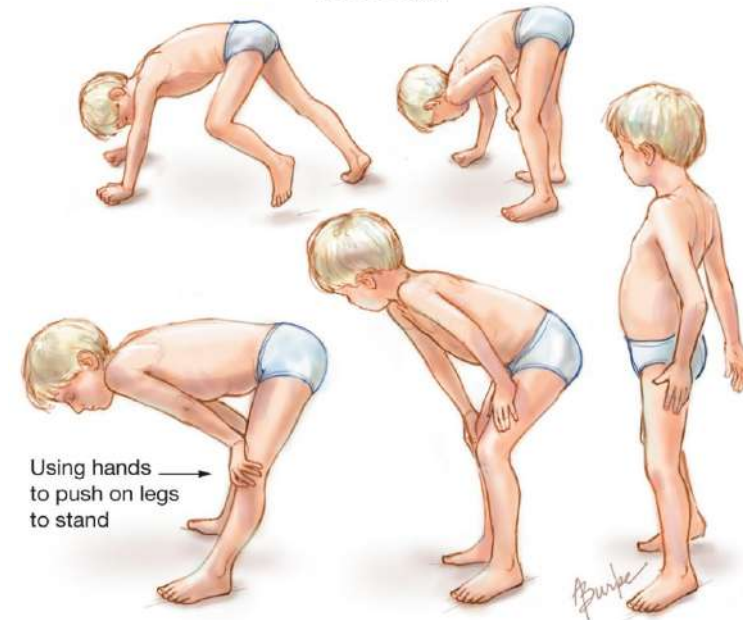
- Progressive muscular weakness and wasting (proximal>distal), with +ve Gowers sign and calf pseudohypertrophy
- Wheelchair-dependent by adolescence (~ age 12)
- Dilated cardiomyopathy (DCM)

**Clinical tests:**

- Elevated creatine kinase: unreliable test for carrier females, tends to decrease with age



Gowers Sign



# Duchenne Muscular Dystrophy

## Molecular tests:

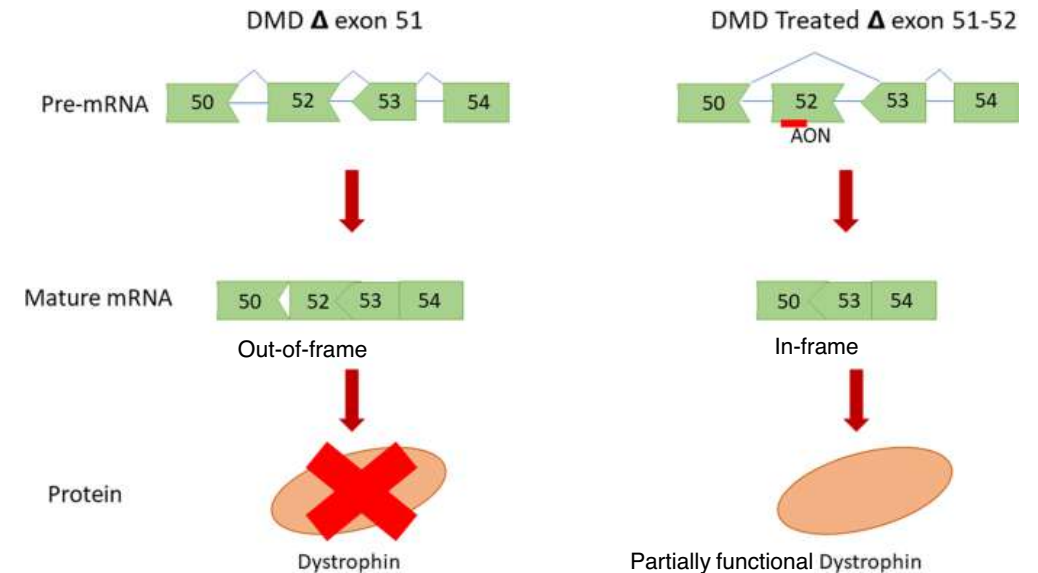
- Multiplex PCR/MLPA: *DMD* gene deletion (65-80% DMD)
- *DMD* sequencing for point mutations (30% DMD)

## Disease mechanism:

- Dystrophin binds actin and other membrane proteins
- Mutations that lead to lack of dystrophin expression

## Management:

- Symptomatic, corticosteroids for muscle strength, dystrophin restoration therapies (exon skipping) for specific DMD variants.



Mechanism of exon skipping therapy for (DMD).

- A deletion of exon 51, leads to premature termination of the transcript, and therefore, no dystrophin production.
- Using an antisense oligonucleotide (AON in red) allows for the skipping of exon 52, resorting to the reading frame, and thus the production of truncated dystrophin protein.

Dzierlega, K., & Yokota, T. (2020). Optimization of antisense-mediated exon skipping for Duchenne muscular dystrophy. *Gene therapy*, 27(9), 407-416.

# Chromosomal Abnormalities

- ? Account for approximately 6% of all recognized congenital abnormalities
- ? Can be categorized as either numerical (trisomy, or monosomy,) or structural (duplication, deletion)
- ? Chromosomal abnormalities can occur when the egg or the sperm is formed or during the early developmental stages of the fetus
- ? Chromosomal imbalance can cause significant structural, developmental and growth problems, which may lead to early miscarriage.
- ? Could be related to dosage effects of the individual genes involved (“additive” model) or general developmental instability caused by a large number of abnormal developmental gene products (“interactive” model).



# Chromosomal Abnormalities

- Numerical chromosomal abnormalities/aneuploidies:
  - Down syndrome (Trisomy 21) *covered in dysmorphology lecture*
  - Patau syndrome (Trisomy 13) *covered in dysmorphology lecture*
- Sex chromosomal abnormalities:
  - Turner syndrome (45,XO) *covered in dysmorphology lecture*
  - **Klinefelter syndrome (47,XXY)**
- Structural chromosomal abnormalities:
  - **Cri du Chat syndrome (5p- syndrome)**
  - **22q11 deletion syndrome**
- Imprinting disorders:
  - Prader-Willi syndrome and Angelman syndrome *covered in Non-Mendelian inheritance lecture*
  - **Beckwith-Wiedemann syndrome and Russel-Silver syndrome**



# Klinefelter Syndrome (47,XXY)

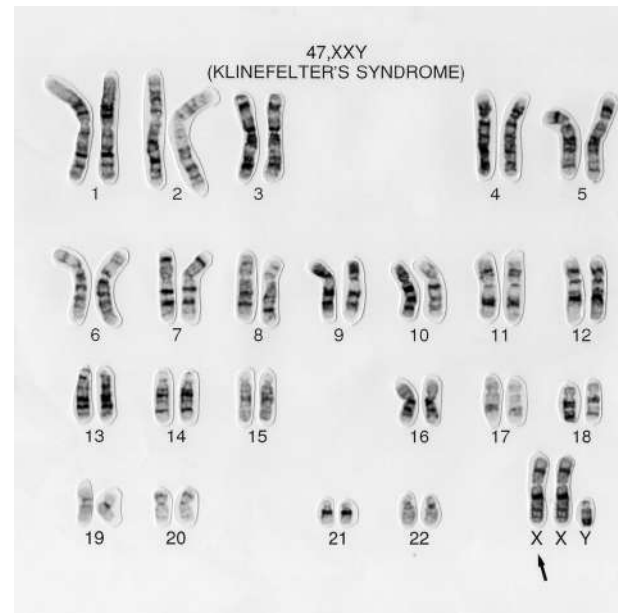
## Clinical features:

- Tall stature
- Slightly delayed motor, language skills, learning difficulties, mild ID
- Infertility, testicular atrophy (small fibrosed testes), azoospermia
- Gynecomastia, female distribution of fat and body hair
- Increased risk of male breast cancer and osteoporosis

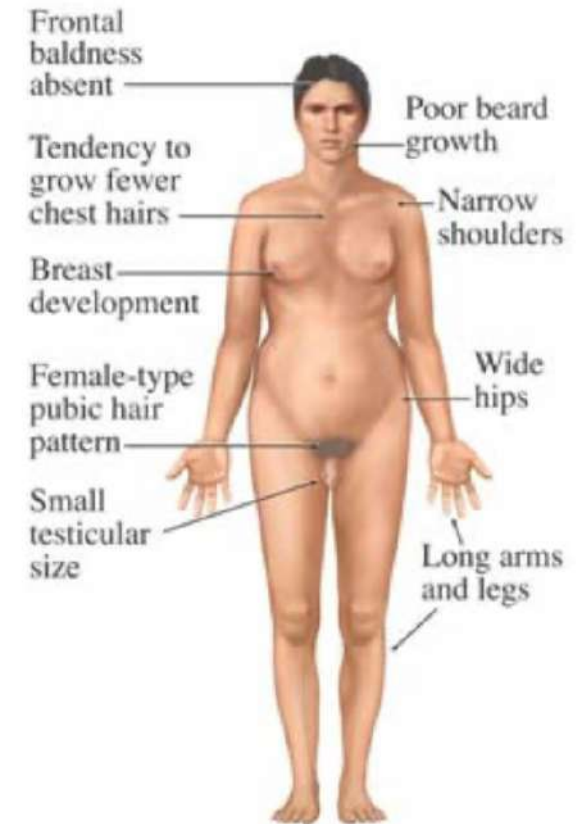
**Karyotype:** shows an additional X chromosome

**Treatment:** Testosterone starting from puberty is beneficial for:

- Development of secondary sexual characteristics
- Long-term prevention of osteoporosis



Credit: [Wessex Reg. Genetics Centre](#). [Attribution 4.0 International \(CC BY 4.0\)](#)



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# Cri-du-Chat (5p minus) Syndrome

**Cytogenetic locus:** 5p15.2

- Terminal deletions of the short arm of chromosomes 5

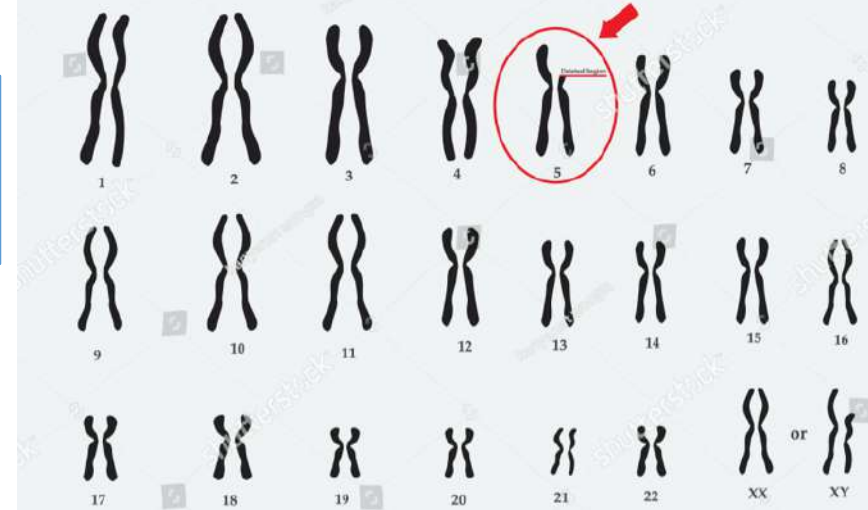
## Clinical features:

- Characteristic cat-like cry of affected neonates due to underdevelopment of the larynx
- Growth deficiency
- Developmental delays, intellectual disability
- Microcephaly, hypotonia
- Dysmorphic features: round face, hypertelorism, micrognathia, epicanthal folds and low-set ears

## Inheritance:

- Mostly → sporadic *de novo* deletions
- 10% → have a parent with balanced translocation

Cri-Du-Chat syndrome or Cat-Cry syndrome



# 22q11.2 Deletion Syndrome (DiGeorge, Velocardiofacial syndrome)

**Responsible genes:** *TBX1* (encodes for T box 1 transcription factor C protein)

**Cytogenetic locus:** 22q11.2

**Inheritance:** AD; > 90% de novo

**Clinical Features:** Characterized by

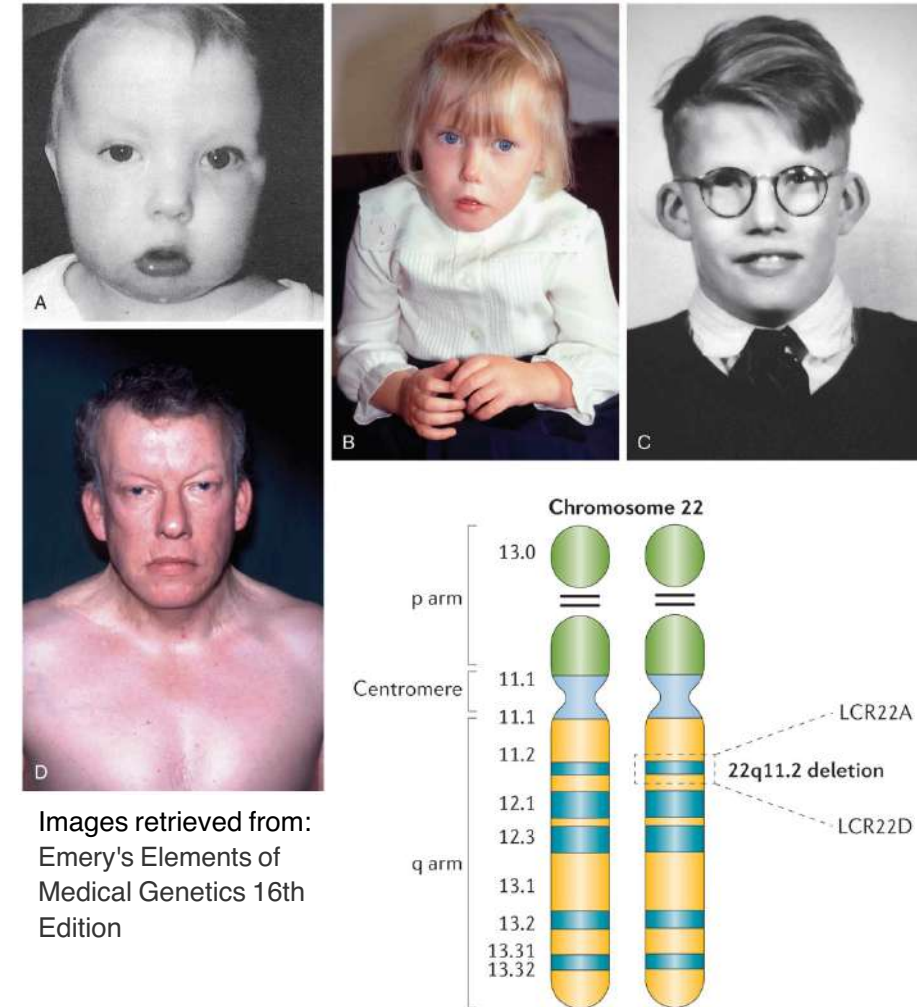
- Heart malformations (particularly involving the cardiac outflow tract/conotruncal defects), and palate abnormalities
- Thymic and parathyroid hypoplasia, and immune dysfunction
- Short stature, DD/learning difficulties, and typical facies
- Psychiatric illness (schizophrenia 25% of adult, autism 20% of children)

**Clinical Tests:** Hypocalcemia, hypoparathyroidism, immunodeficiency, and growth hormone deficiency

**Molecular Tests:** FISH, microarray, or MLPA. 3-Mb deletion most common

**Disease Mechanism:** Abnormal development of the pharyngeal arches related to haploinsufficiency of *TBX1* gene

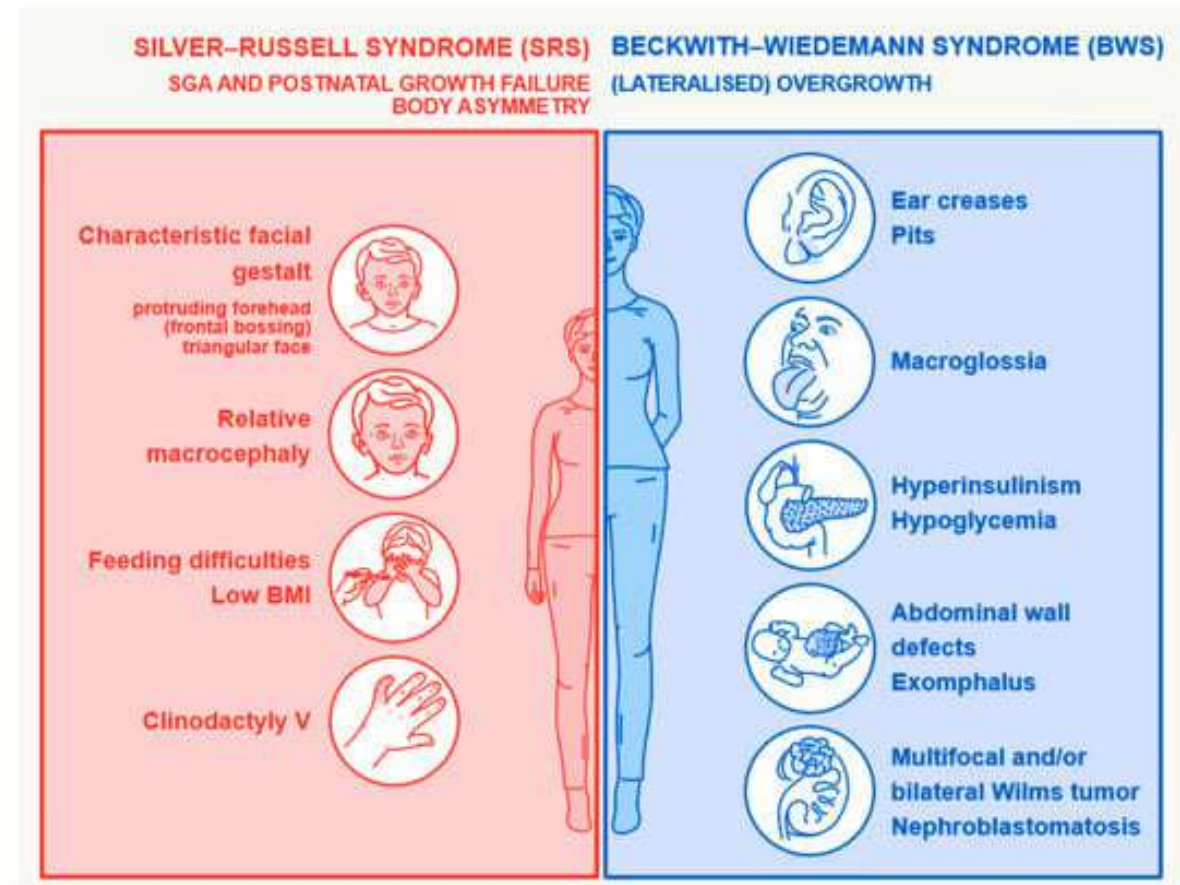
**Treatment:** symptomatic, no live vaccines if immunodeficient



# Beckwith-Wiedemann Syndrome (BWS)

## Silver-Russell Syndrome (SRS or RSS)

- BWS and SRS are 2 clinically opposite growth-affecting congenital imprinting disorders
- The expression of both syndromes usually depends on the parental origin of the chromosome in which the imprinted genes reside
- Caused by disruptions in gene regulation/expression of imprinted genes including growth factors and tumor suppressor genes on chromosome 11p15
- Leads overgrowth in BWS and growth restriction in SRS



Wessler, K., Kraft, F., & Eggermann, T. (2019). Molecular and clinical opposite findings in 11p15. 5 associated imprinting disorders: characterization of basic mechanisms to improve clinical management. *International Journal of Molecular Sciences*, 20(17), 4219.



# Beckwith-Wiedemann Syndrome (BWS)

Cytogenetic locus: 11p15.5

## Clinical Features:

- **Overgrowth Syndrome**; characterized by macrosomia (prenatal and/or postnatal overgrowth), macroglossia (large tongue), abdominal wall defect (omphalocele, umbilical hernia, diastasis recti), and neonatal hypoglycemia
- **Other Features**: hemihyperplasia, visceromegaly, renal abnormalities, ear anomalies (anterior earlobe creases, posterior helical pits), cleft palate, and increased risk of embryonal tumors (Wilms tumor, hepatoblastoma)
- Associated with assisted reproductive technologies (IVF)

## Molecular Tests:

- Methylation analysis
- Deletion
- UPD



Images retrieved from: Emery's Elements of Medical Genetics 16th Edition



# Silver-Russel Syndrome (SRS or RSS)

## Cytogenetic locus:

- Chromosome 11p15.5
- Chromosome 7

## Clinical Features

- Marked prenatal and postnatal growth restriction (opposite of BWS overgrowth)
- Normal head size with small, triangular face (relative macrocephaly)
- Body asymmetry, fifth finger clinodactyly

## Molecular Test:

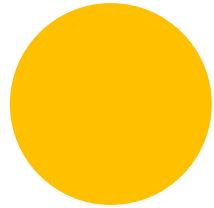
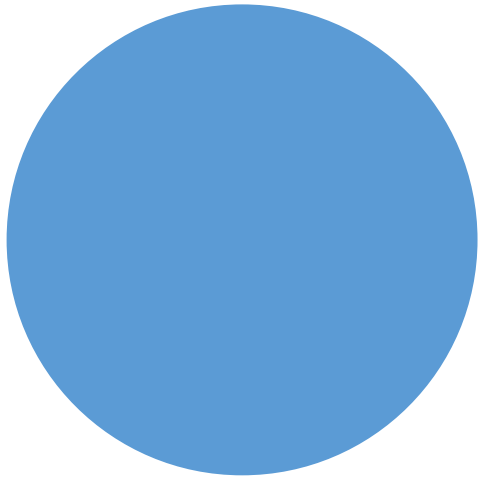
- Methylation analysis
- Deletion
- UPD
- Maternal UPD chr 7 (~10%)



# References

- Emery's Elements of Medical Genetics 16th Edition
- Oxford Desk Reference Clinical Genetics by Helen V. Firth and Jane A. Hurst
- [GeneReviews, © 1993-2023 University of Washington](#)





Thank You!  
Questions/Discussions?!

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